Complete and Invariant Instance Classifier Refinement for Weakly Supervised Object Detection in Remote Sensing Images

Xiaoliang Qian, *Member, IEEE*, Chenhao Wang, Wei Wang, Xiwen Yao, *Member, IEEE*, and Gong Cheng, *Member, IEEE*

***Abstract*— Weakly supervised object detection (WSOD) in remote sensing images (RSIs) is used to detect high-value objects by utilizing image-level labels. However, the current models still have two problems. First, the misclassification of neighboring instances has easily occurred because the one-hot label is assigned to all of the seed instances and their neighboring instances. Second, the supervisory information of each instance classifier refinement (ICR) branch is generated from the predicted class score of the upper ICR branch rather than the real label; thus, the prediction mistake of each ICR branch will be accumulated with the propagation of supervisory information. To address the first problem, a complete definition of the pseudosoft label (CPSL) of instances is proposed to directly train each ICR branch, where the CPSL of seed instances is defined according to the predicted class scores of upper ICR branch, and the CPSL of other instances is determined by the spatial distance weighted feature similarity (FS) between them and seed instances. To handle the second problem, an invariant multiple instance learning (IMIL) scheme is proposed to indirectly train each ICR branch by using real image-level labels. Furthermore, the affine transformations of the original image are incorporated into the baseline model to enhance the invariance of our model. The ablation studies verify the effectiveness of CPSL, IMIL, and their combination. The quantitative comparisons with popular methods show that 73.63% (31.08%) mean average precision (mAP) and 79.88% (57.52%) correct localization (CorLoc) of our method are the best on the NWPU VHR-10.v2 (DIOR) dataset, and the qualitative comparisons intuitively demonstrate it again.**

***Index Terms*— Complete definition of pseudosoft label (CPSL), invariant multiple instance learning (IMIL), remote sensing image (RSI), weakly supervised object detection (WSOD).**

I. Introduction

WEAKLY supervised object detection (WSOD) accomplishes object classification and localization by training the object detector solely using image-level labels, which saves a lot of time and labor cost compared with fully supervised object detection [1], [2], [3], [4], [5], [6], [7] (FSOD), which utilizes instance-level labels. WSOD can be used to detect high-value objects in remote sensing images (RSIs), providing technical support for urban planning [8], [9], [10], [11], military strategy [12], [13], [14], and other applications [15].

The weakly supervised deep detection networks (WSDDNs) [16], as a milestone work, first combined deep learning with multiple instance learning (MIL) to train the object detector by using image-level labels; the trained object detector predicts the class scores of proposals to determine whether or not the proposals are objects and their classes. Afterward, the online instance classifier refinement (OICR) [17] networks made significant improvements on the basis of WSDDN [16], which introduced a number of instance classifier refinement (ICR) branches to iteratively refine the class scores of proposals. Specifically, the proposal with the highest class score inferred from the current ICR branch is regarded as the seed instance of each class, and the seed instances with their neighboring instances are assigned one-hot labels to supervise the next ICR branch.

Up until now, the majority of WSOD methods [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32] made improvements on the basis of OICR, which can be broadly classified into two classes. The first class methods aimed to address the issue that OICR only selected the proposal with the highest class score as seed instance, which was difficult to detect multiple objects with the same class such that the multiple instance self-training (MIST) method [28] selected multiple instances with top class score as candidate seed instances; then, the final seed instances were determined through local nonmaximum suppression (NMS) [33] among candidate seed instances. Other congener methods include proposal cluster learning (PCL) [32], weakly supervised contrastive loss (WSCL) [22], and so on. Considering the arbitrary object orientation and scale changes in RSIs, the second category methods aimed to construct an affine invariant WSOD model. For example, the invariant and equivariant network (RINet) [23] generated different affine transformation features that were imported into different ICR branches and learned a WSOD model that was invariant to various affine transformations through the supervising between neighboring ICR branches.

Manuscript received 23 April 2024; accepted 24 May 2024. Date of publication 3 June 2024; date of current version 18 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62076223, in part by the Key Research Project of Henan Province Universities under Grant 24ZX005, and in part by the Key Science and Technology Program of Henan Province under Grant 232102211018. *(Corresponding authors: Wei Wang; Gong Cheng.)*

Xiaoliang Qian, Chenhao Wang, and Wei Wang are with the College of Electrical and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou 450002, China (e-mail: qxl_sunshine@163.com; star_chenhao@163.com; wangwei-zzuli@zzuli.edu.cn).

Xiwen Yao and Gong Cheng are with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: yaoxiwen@nwpu.edu.cn; gcheng@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3408895

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

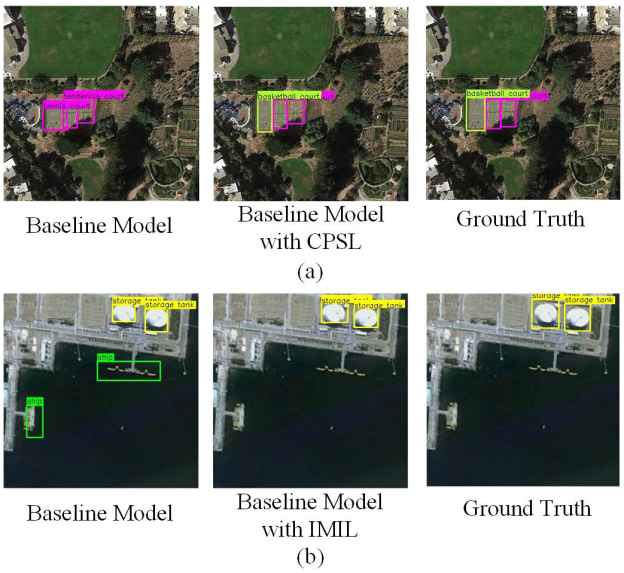


Fig. 1. Illustration of the motivations and contributions of (a) CPSL and (b) IMIL.

Other congener methods include comprehensive attention self-distillation (CASD) [31] and so on. Although the above methods have made some breakthroughs in WSOD, some common issues should be addressed.

For the first issue, to the best of our knowledge, the current methods assign a one-hot label to all seed instances and their neighboring instances, which assumes that the probability that each seed instance belongs to the ground-truth object is equal and the neighboring instances have the same characteristic as seed instances; it is obvious that it is not true. Consequently, the misclassification of neighboring instances usually occurs because the pseudolabels of neighboring instances are coped from the labels of seed instances. As shown in Fig. 1(a), the baseline model (see Section IV-E) mistakenly assigns the label of the tennis court to the basketball court because the basketball court is a neighboring instance of the tennis court.

For the second issue, the pseudolabels of each ICR branch are generated from the class score predicted by the upper ICR branch rather than the real labels. As a result, the prediction mistakes of each ICR branch will be accumulated with the propagation of supervisory information. Consequently, the misclassification usually occurs because of the accumulated errors. As shown in Fig. 1(b), the baseline model mistakenly identifies some complex background regions as the ships that have not existed in the RSI.

To handle the first issue, this article proposes a complete definition of the pseudosoft label (CPSL) of each instance, and the novel CPSL is considered the first level supervision information to directly train each ICR branch. First, the soft label of each seed instance is defined by using the class score inferred from the upper ICR branch to distinguish the importance of each seed instance. Afterward, the soft labels of neighboring instances are defined according to the spatial distance weighted feature similarity (FS) between neighboring instances and their corresponding seed instances. The definition of background instances is similar to the definition of neighboring instances. As shown in Fig. 1(a), the baseline model with the proposed CPSL can avoid the misclassification of the basketball court because the difference between the tennis court and neighboring basketball court can be precisely reflected by using the proposed CPSL.

To handle the second issue, this article proposes an invariant MIL (IMIL) scheme. Specifically, the real image-level labels are considered the second-level supervision information to indirectly train each ICR branch through MIL. In other words, the image-level annotations and the aforementioned CPSL are jointly used to train each ICR branch. Consequently, the prediction mistake of each ICR branch can be suppressed as much as possible by using more complete supervisory information, and the misclassification problem will be alleviated to some extent, especially for the object categories not exist in the RSI. As shown in Fig. 1(b), the misclassification of ships is avoided because our model knows that the ships have not existed in the RSI through the proposed IMIL scheme.

In addition, inspired by Feng et al. [23], the common affine transformations are also incorporated into our method to learn a WSOD model, which is invariant to the rotation, flipping, and scaling of objects in RSIs. In summary, the proposed CPSL and IMIL are used to support the complete and invariant ICR (CICR) through direct and indirect training supervised by the aforementioned CPSL and real image labels, respectively.

The contributions of this article are summarized in the following.

1) A CPSL of each instance is proposed to address the misclassification of neighboring instances caused by assigning a one-hot label to all of the seed instances and their neighboring instances. The CPSL of each seed instance is defined by using the class score inferred from the upper ICR stream, and the CPSL of neighboring instances and background instances is defined by the spatial distance weighted FS between them and their corresponding seed instances. The proposed CPSL is used to directly train each ICR branch.
2) An IMIL scheme is proposed to suppress the prediction mistake of each ICR branch caused by only using the prediction results of the upper ICR branch to derive the supervisory information of the current ICR branch. The ground-truth image-level labels are combined with the aforementioned CPSL to construct more complete supervisory information of each ICR branch and are used to indirectly train each ICR branch through MIL.

II. Related Work

This section includes two parts of content. First, the baseline WSOD model adopted in this article is introduced, which includes the WSDDN [16], OICR [17], and multiple bounding box regression (BBR) branches. Second, the other WSOD methods except the ones introduced in Section I are briefly described.

A. Baseline WSOD Model

1) Weakly Supervised Deep Detection Network: First, the proposal set of input image I, denoted as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

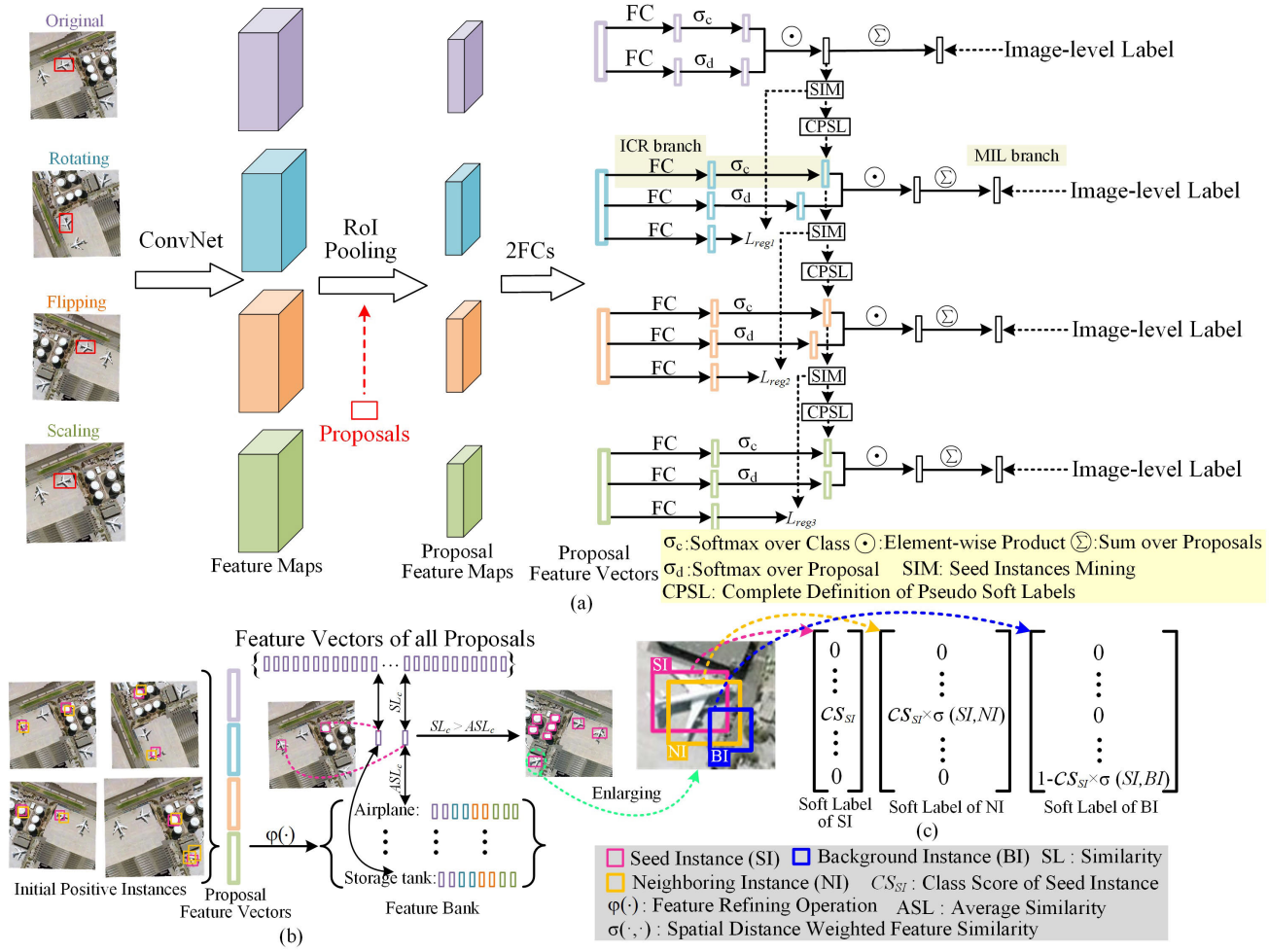


Fig. 2. Illustration of our method. (a) Overall framework of our method. (b) Illustration of seed instance mining (SIM) module. (c) Illustration of CPSL. The captions in (a) are shown in the buff region, and the captions in (b) and (c) are shown in the gray region.

$P = \{p_1, \ldots, p_m, \ldots, p_M\}$, is generated by using the selective search (SS) [34], where M represents the quantity of elements in the set P. Then, the feature vectors of all proposals, denoted as $\{f_m \in \mathbb{R}^{4096}\}_{m=1}^{M}$, are obtained by importing I and P into the backbone network, region of interest (RoI) pooling layer, and two fully connected (FC) layers. Afterward, the two matrices, denoted as $Z_c, Z_d \in \mathbb{R}^{C\times M}$, are obtained by importing the $\{f_m\}_{m=1}^{M}$ into two parallel FC layers, where C represents the number of categories. Z_c (Z_d) contains the class (proposal) prediction scores for each proposal (class) [16]. The score matrix of P, denoted as $Z \in \mathbb{R}^{C\times M}$, is obtained as follows:

$$Z = \sigma_c(Z_c) \odot \sigma_d(Z_d) \tag{1}$$

where $\odot$ denotes elementwise product, and σ_c and σ_d denote softmax over class and proposal, respectively. The image-level class score of I is obtained by the following equation:

$$\phi_c = \sum_{m=1}^{M} z_{c,m} \tag{2}$$

where ϕ_c denotes the class score of I in the cth class and $z_{c,m}$ represents the element in the cth row and mth column of Z. The binary cross-entropy (BCE) loss of WSDDN, denoted as L_B, is obtained as follows:

$$L_B = -\sum_{c=1}^{C} (y_c \log \phi_c + (1 - y_c) \log(1 - \phi_c)) \tag{3}$$

where $y_c = 1$ or 0 represents whether or not I contains class c.

2) Online Instance Classifier Refinement: The K ICR streams are added into the WSDDN [16] to iteratively refine the class score of each proposal, where each ICR stream includes an FC layer and a softmax classifier. The class score matrix of kth ICR branch, denoted as $\mathrm{CS}^k \in \mathbb{R}^{(C+1)\times M}$, is obtained by importing the $\{f_m\}_{m=1}^{M}$ into the ICR branch, where the $(C+1)$th row of CS^k represents the background class. The cross-entropy (CE) loss L_{cls}^k of the kth ICR stream is obtained as follows:

$$L_{\mathrm{cls}}^k = -\sum_{m=1}^{M}\sum_{c=1}^{C+1} w_m^k y_{c,m}^k \mathrm{cs}_{c,m}^k \tag{4}$$

where $\mathrm{cs}_{c,m}^k \in \mathrm{CS}^k$ denotes the class score of kth ICR branch of mth proposal in the class c, $w_m^k = \mathrm{cs}_{c,m_c^{k-1}}^{k-1}$ denotes the weight of mth proposal, $\mathrm{cs}_{c,m_c^{k-1}}^{k-1}$ denotes the highest class score in the kth ICR branch, m_c^{k-1}th proposal denotes the seed instane of the kth ICR branch in class c, $y_{c,m}^k$ denotes the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

pseudolabel of and mth proposal, and $y_{c,m}^k = 1$ if $m = m_c^{k-1}$ or the mth proposal is the neighboring proposal of the m_c^{k-1}th proposal; otherwise, $y_{c,m}^k = 0$.

3) Bounding Box Regression: Inspired by Yao et al. [21], Seo et al. [22], Ren et al. [28], Chen et al. [35], and Zeng et al. [36], the K BBR branches are added to OICR framework for improving the localization performance. The loss of the kth BBR branch, denoted as L_{reg}^k, is given as follows:

$$L_{\text{reg}}^k = \frac{1}{|R^k|} \sum_{r=1}^{|R^k|} \left(\text{smooth}_{L1}\left(t_r, \hat{t}_r\right)\right) \tag{5}$$

where R_k denotes the set of positive instances in the kth ICR branch, $|\cdot|$ denotes the cardinality operation, $\text{smooth}_{l1}(\cdot)$ denotes the smooth $L1$ function [3], and t_r and $\hat{t}_r$ denote the prediction offset and target offset of the rth positive instance, respectively.

B. Other WSOD Methods

To handle the occasion that an image contains multiple objects with the same class, the current WSOD methods can be broadly classified into two categories according to the mining strategy of seed instances.

The first kind of method only uses the class score of each proposal to mine seed instances. For example, PCL [32] divided the proposals into multiple clusters according to their class score, and the proposals belonging to the cluster with the highest class score are selected as the seed instances. Wu et al. [37] proposed an adaptive instance mining strategy that used the distribution of class scores to adaptively mine seed instances. RINet [23] considers the instances with the highest class score as the seed instances in each affine transformation branch, and the seed instances of each branch are coupled to obtain the final seed instances. Other similar methods include CASD [31], dynamic curriculum learning (DCL) [21], multiple detection branches framework [38], mask-guided self-correction method [39], multiscale image splitting based feature enhancement module [19], and so on.

The second kind of method jointly uses the class score of each proposal and other metrics to mine seed instances. For example, the pseudoinstance soft labels mining (PISLM) model [18] proposed a novel metric named proposal quality score to mine seed instances, and the metric is obtained by combining the dual-context projection score with the class score. Wang et al. [27] proposed a multiple instance graph (MIG) learning framework that calculated the similarity between each proposal and representative instances to mine seed instances. A novel objectness score was proposed by objectness score and difficulty evaluation score (OD-DES) [40] to mine the seed instances, which was obtained by fusing the object completeness score and class score. Other similar methods include WSCL [22], the object instance mining model [41], the semantic segmentation-guided pseudolabel mining and instance re-detection (SGPLM) model [20], the coupled multiple instance detection network [42], and so on.

III. Proposed Method

A. Seed Instances' Mining With Affine Invariance

As mentioned before, OICR assigned the instance with the highest class score as a seed instance, which missed many objects with the same class. To discover more valid seed instances, inspired by WSCL [22], a novel SIM module is proposed in this article. The SIM model constructs a feature bank for all object classes by using affine transformation features; afterward, the instance in which the similarity between its features and feature bank is larger than an adaptive threshold is selected as a seed instance. The SIM model can mine more valid seed instances with the characteristic of affine invariance in this way.

As shown in Fig. 2(b), first, the initial positive instances of original, rotating, flipping, and scaling images in class c are mined through the mining strategy of OICR [17], where the assemble of them is denoted as PI^c, and each of them is denoted as $\text{pi}_j^c \in \text{PI}^c$. Second, the refined feature vector of pi_j^c, denoted as $\text{bf}_j^c \in \mathbb{R}^{128}$, is extracted through following equation:

$$\text{bf}_j^c = \left\{\varphi\left(\text{pf}_j^c\right) \middle| j \in \left[1, \left|\text{PI}^c\right|\right]\right\} \tag{6}$$

where $\text{pf}_j^c \in \mathbb{R}^{4096}$ denotes the initial feature vector of pi_j^c; 4096 denotes the dimensions of proposal feature vector obtained through ConvNet, RoI pooling operation, and two FCs, as shown in Fig. 2(a); and $\varphi(\cdot)$ denotes feature refining operation, which includes two FCs layer and a normalization operation. The feature bank of class c, denoted as BF^c, is, thus, defined as $\text{BF}^c = \{\text{bf}_j^c\}_{j=1}^{|\text{PI}^c|}$.

Third, the similarity threshold SLT_k^c that determines whether or not a proposal belongs to the seed instance of class c in the kth ICR stream is obtained as follows:

$$\text{SLT}_k^c = \frac{1}{|\text{BF}^c|} \sum_{j=1}^{|\text{BF}^c|} \text{sim}\left(\varphi\left(f_{m_c^{k-1}}\right), \text{bf}_j^c\right) \tag{7}$$

where $\text{sim}(\cdot)$ denotes the similarity calculation that is implemented through the inner product and $f_{m_c^{k-1}}$ represents the feature vector of the m_c^{k-1}th proposal that has the highest class score of category c in the $(k-1)$th ICR stream. Then, a similarity $\text{SL}_{c,m}^k$ of the mth proposal of category c in the kth ICR stream is defined as follows:

$$\text{SL}_{c,m}^k = \text{sim}\left(\varphi(f_m), \varphi\left(f_{m_c^{k-1}}\right)\right). \tag{8}$$

The mth proposal is regarded as the seed instance of category c in kth ICR stream if $\text{SL}_{c,m}^k > \text{SLT}_c^k$, and the assemble of all seed instances of category c in the kth ICR stream is denoted as SI_c^k.

On the whole, the proposed SIM module is similar to the support-set-guided pseudolabeling approach [43]. Specifically, the feature vectors of representative instances are used to construct feature banks, which is similar to the support-set [43], and the seed instances are mined according to the similarity between the feature vectors of instances and feature bank.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

B. Complete Definition of Pseudosoft Label for First Level Supervision of ICR Branches

As shown in Fig. 2(c), this article replaces the pseudolabel of each seed instance from one-hot label to CPSL defined by the predicted class score of the upper ICR branch. It is obvious that the importance of each seed instance can be distinguished by CPSL. The CPSL of neighboring and background instances is defined according to the spatial distance weighted FS between them and corresponding seed instances. Compared with coping the label of seed instance, the CPSL gives a more precise label for neighboring and background instances, which can alleviate the misclassification of neighboring and background instances as much as possible. The CPSL of all instances is used to directly train K ICR branches

$$\mathrm{sy}_{c,m}^{k}=\begin{cases}\mathrm{cs}_{c,m}^{k-1}, & \text{if } p_m\in \mathrm{SI}_c^k\\ \mathrm{ncs}_c^{k-1}\times\sigma\left(p_m,\mathrm{nsi}_c^k\right), & \text{if } p_m\in \mathrm{NI}_c^k\\ 1-\mathrm{bcs}_c^{k-1}\times\sigma\left(p_m,\mathrm{bsi}_c^k\right), & \text{if } p_m\in \mathrm{BI}_c^k\end{cases}\tag{9}$$

where p_m denotes the mth proposal, $\mathrm{sy}_{c,m}^k$ denotes the CPSL of p_m of category c in the kth ICR stream, NI_c^k represents the assemble of neighboring instances of SI_c^k, BI_c^k denotes the assemble of background instances, $\mathrm{cs}_{c,m}^{k-1}$ denotes the predicted class score of p_m of category c in the $(k-1)$th ICR stream, $\mathrm{nsi}_c^k\in\mathrm{SI}_c^k$ represents the seed instance of p_m if p_m is the neighboring instance of nsi_c^k, ncs_c^{k-1} denotes the predicted class score of nsi_c^k in the $(k-1)$th ICR stream, $\mathrm{bsi}_c^k\in\mathrm{SI}_c^k$ represents the nearest seed instance of p_m if $p_m\in\mathrm{BI}_c^k$, bcs_c^{k-1} denotes the predicted class score of bsi_c^k in the $(k-1)$th ICR stream, and $\sigma(\cdot,\cdot)$ indicates the spatial distance weighted FS between p_m and seed instance and is defined as follows:

$$\sigma\left(p_m,\mathrm{si}_c^k\right)=\mathrm{IoU}\left(p_m,\mathrm{si}_c^k\right)\times\mathrm{sim}\left(f_{p_m},f_{\mathrm{si}_c^k}\right)\tag{10}$$

where si_c^k denotes any seed instance in SI_c^k, $\mathrm{IoU}(p_m,\mathrm{si}_c^k)$ denotes intersection over union between p_m and si_c^k, f_{p_m} and $f_{\mathrm{si}_c^k}$ denote the feature vector of p_m and si_c^k, respectively, $\mathrm{sim}(f_{p_m},f_{\mathrm{si}_c^k})$ denotes the similarity between f_{p_m} and $f_{\mathrm{si}_c^k}$, and the meaning of $\mathrm{sim}(\cdot,\cdot)$ has been explained in (7).

Hence, the first level supervision for kth ICR stream can be implemented as follows:

$$L_{\mathrm{cls}}^k=-\sum_{m=1}^{M}\sum_{c=1}^{C+1}w_m^k\mathrm{sy}_{c,m}^k\mathrm{cs}_{c,m}^k\tag{11}$$

where L_{cls}^k represents the first level loss of the kth ICR branch, $\mathrm{cs}_{c,m}^k$ represents the predicted class score of p_m of category c in the kth ICR stream, C denotes the number of classes, and w_m^k denotes the weight of p_m in the k ICR branch [17].

C. Invariant MIL for Second-Level Supervision of ICR Branches

As shown in Fig. 2(a), to suppress the accumulated prediction mistake of K ICR branches, the proposed IMIL scheme uses the ground-truth image-level label to indirectly train K ICR branches. The ground-truth image-level label can be considered the second-level supervision of ICR branches, and it is invariant with the affine transformation of input RSI. Apparently, the misclassification of categories that do not exist in RSI can be avoided by adding the supervision of image-level labels.

First, the proposal feature vector $\{f_m\}_{m=1}^M$ is imported into the kth assistant branch to obtain assistant score matrix $\mathrm{AS}^k\in\mathbb{R}^{(C+1)\times M}$, where each assistant branch includes an FC layer and σ_d. Afterward, the image-level class score of cth class in the kth MIL branch, denoted as μ_c^k, is obtained as follows:

$$S^k=\mathrm{AS}^k\odot\mathrm{CS}^k\tag{12}$$

$$\mu_c^k=\sum_{m=1}^{M}s_{c,m}^k\tag{13}$$

where S^k denotes the score matrix of the kth MIL branch and $s_{c,m}^k\in S^k$ denotes the element in the cth row and the mth column of S^k. The loss of kth MIL branch, denotes as L_{MIL}^k, is obtained as follows:

$$L_{\mathrm{MIL}}^k=-\sum_{c=1}^{C}\left(y_c\log\mu_c^k+(1-y_c)\log\left(1-\mu_c^k\right)\right).\tag{14}$$

It is worth noting that the training of K MIL branches has the characteristic of affine invariance since y_c is invariant for K MIL branches of which the corresponding input images are affine transformation versions of the original image.

D. Overall Training Loss

The total training loss L of our CICR model is obtained as follows:

$$L=\mathrm{IDS}(I)\times\left(L_B+\sum_{k=1}^{K}\left(L_{\mathrm{cls}}^k+L_{\mathrm{reg}}^k+L_{\mathrm{MIL}}^k\right)\right)\tag{15}$$

where $\mathrm{IDS}(I)$ denotes the image-level difficulty score (IDS) of input image I, which is inferred by the difficulty estimator module [44].

In the inference stage, the proposal feature vectors of the original image are fed into the K trained ICR streams and the K trained BBR branches to jointly infer the detection results.

IV. Experiments

A. Datasets and Evaluation Metrics

Our method is validated on two publicly RSI benchmarks, i.e., NWPU VHR-10.v2 [45], [46] and DIOR [47]. The NWPU VHR-10.v2 dataset includes 1172 images with the size of 400 × 400 pixels, which contains ten classes with 2775 object instances. The training set (679 images) and the validation set (200 images) are used for training our CICR model, and the testing set (293 images) is used for evaluating the capability of our CICR model. The DIOR dataset includes 23 463 images with the size of 800 × 800 pixels, which contains 20 classes with 192 472 object instances. The training set (5862 images) and the validation set (5863 images) are used for training, and the testing set (11 738 images) is used for testing.

The mean average precision (mAP) is employed to measure detection accuracy in the testing set, and the correct localization (CorLoc) [48] is used to measure localization capability in the training and validation sets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

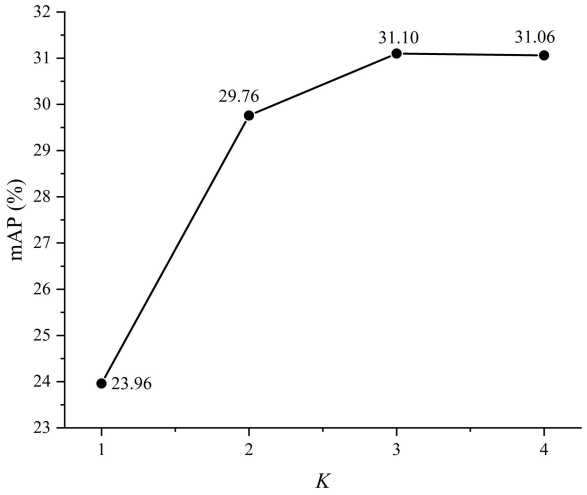


Fig. 3. Parameter analysis of K in terms of mAP on the DIOR dataset.

TABLE I
ABLATION STUDY OF FS ON THE DIOR DATASET

Method	mAP (%)	CorLoc (%)
Baseline + CPSL without FS	23.92	47.13
Baseline + CPSL with FS	24.59	48.23

TABLE II
ABLATION STUDY OF CPSL AND CMIL ON THE DIOR DATASET

Method	mAP (%)	CorLoc (%)
Baseline with One-hot Label	20.10	42.79
Baseline + CPSL	24.59	48.23
Baseline + IMIL	27.11	51.91
Baseline + CPSL + IMIL (Ours)	31.06	57.75

B. Implementation Details

The stochastic gradient descent algorithm is used to optimize our model. The VGG16 [49] pretrained on the ImageNet dataset [50] is used as the backbone of our CICR model. The NMS [33] threshold in the inference stage is set to 0.3, and the quantity of ICR branches is set to 3, i.e., $K = 3$.

The two datasets are augmented through horizontal flipping and rotation with 90° and 180° [18], [19], [20], [21], [23]. The momentum, weight decay, and batch size are set to 0.9, 0.005, and 8, respectively. The number of iterations is set to 30 K (60 K) for the NWPU VHR-10.v2 (DIOR) dataset. The learning rate is first set to 0.1, and it is reduced to 10% of the previous value at 20 K (50 K) and 26 K (56 K) for the NWPU VHR-10.v2 (DIOR) dataset, respectively.

Our source code runs on the Ubuntu 16.04 platform with eight GPUs (Titan RTX) and is implemented in the PyTorch framework.

C. Overview Execution

As shown in Fig. 2(a), first, the original image, randomly rotated image (0°, 90°, 180°, and 270°), randomly flipped image (no flipping, horizontal flipping, and vertical flipping), and randomly scaled image (480, 576, 688, 864, 1000, and 1200) are, in turn, fed into the backbone to obtain corresponding feature maps. Second, the corresponding transformed proposals and the feature maps are successively imported into the RoI pooling layer and two FC layers to obtain the feature vectors of all proposals. Then, the feature vectors of the original proposals are imported into the WSDDN model, and the feature vectors of rotated, flipped, and scaled proposals are imported into the first, second, and third MIL and BBR branches, respectively. Third, the seed instances are mined through the SIM module. Fourth, the pseudolabel of each instance is defined by the proposed CPSL, which is considered the first level supervision information to directly train each ICR branch. Finally, to suppress the prediction mistake of each ICR branch, the real image-level labels are considered the second-level supervision information to indirectly train each ICR branch through the proposed IMIL scheme.

In the training stage, the image-level labels are used to supervise the training of the WSDDN model through the BCE loss. The supervision information of each ICR branch includes the CPSL of all instances and real image-level label, as shown in Fig. 2(a), where the CPSL of all instances is used to directly train each ICR branch through the CE loss and the real image-level label is used to train each MIL branch through the BCE loss; consequently, each ICR branch that is a part of each MIL branch can be indirectly trained with the training of MIL branch. The training loss of each ICR branch is the sum of the above CE and BCE loss supervised by the CPSL and image-level label. The locations of seed instances of each ICR branch are used to supervise the training of the attached BBR branch through smooth $L1$ loss. The total loss of the whole network is the sum of the loss of the WSDDN model, K ICR, and BBR branches. In addition, the WSDDN model, K ICR, and BBR branches have a close relationship with each other. Specifically, the seed instances and their CPSL of the first ICR branch are inferred from the predicted class score of the WSDDN model, and the ones of other ICR branches are inferred from the predicted class score of the upper ICR branch. The supervision information of each BBR branch comes from the seed instances of the corresponding ICR branch.

In the inference stage, only the original RSI is imported into our CICR model, and the initial predicted results are obtained by taking the average of three ICR and BBR branches. Finally, the NMS operation is imposed on the initial predicted results to obtain the final detection results.

D. Parameters Analysis

The hyperparameter K that denotes the number of ICR branches is quantitatively analyzed in terms of mAP on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISONS WITH POPULAR WORKS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF MAP (%)

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	mAP
Fast R-CNN [3]	90.91	90.60	89.29	47.32	100.00	85.85	84.86	88.22	80.29	69.84	82.72
Faster R-CNN [1]	90.90	86.30	90.53	98.24	89.72	69.64	100.00	80.11	61.49	78.14	84.51
WSDDN [16]	30.08	41.72	34.98	88.90	12.86	23.85	99.43	13.94	1.92	3.60	35.12
OICR [17]	13.66	67.35	57.16	55.16	13.64	39.66	92.80	0.23	1.84	3.73	34.52
PCL [32]	26.00	63.76	2.50	89.80	64.45	76.07	77.94	0.00	1.30	15.67	39.41
MELM [51]	80.86	69.30	10.48	90.17	12.84	20.14	99.17	17.10	14.17	8.68	42.29
MIST [28]	69.68	49.16	48.55	80.91	27.08	79.85	91.34	46.99	8.29	13.36	51.52
DCL [21]	72.70	74.25	37.05	82.64	36.88	42.27	83.95	39.57	16.82	35.00	52.11
PCIR [30]	90.78	78.81	36.40	90.80	22.64	52.16	88.51	42.36	11.74	35.49	54.97
MIG [27]	88.69	71.61	75.17	94.19	37.45	47.68	100.00	27.27	8.33	9.06	55.95
TCA [29]	89.43	78.18	78.42	90.80	35.27	50.36	90.91	42.44	4.11	28.30	58.82
SAE [24]	82.91	74.47	50.20	96.74	55.66	72.94	100.00	36.46	6.33	31.89	60.72
SPG [26]	90.42	81.00	59.53	92.31	35.64	51.44	99.92	58.71	16.99	42.99	62.80
PISLM [18]	87.60	81.00	57.30	94.00	36.40	80.40	100.00	56.90	9.80	35.60	63.80
SGPLM [20]	90.70	79.90	69.30	97.50	41.60	77.50	100.00	44.40	17.20	33.50	65.20
RINet [23]	90.30	86.30	79.60	90.70	58.20	80.40	100.00	57.70	18.90	41.60	70.40
AE-IS [25]	91.00	88.20	78.30	93.20	60.60	82.40	100.00	60.40	19.60	45.80	72.00
CICR(ours)	86.83	81.79	72.86	91.84	75.19	85.86	100.00	69.69	10.96	62.98	73.80

TABLE V
COMPARISONS WITH POPULAR WORKS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF CORLOC (%)

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	CorLoc
WSDDN [16]	22.32	36.81	39.95	92.48	17.96	24.24	99.26	14.83	1.69	2.89	35.24
OICR [17]	29.41	83.33	20.51	81.76	40.85	32.08	86.60	7.41	3.70	14.44	40.01
PCL [32]	11.76	50.00	12.82	98.65	84.51	77.36	90.72	0.00	9.26	15.56	45.06
MELM [51]	85.96	77.42	21.43	98.33	10.71	43.48	95.00	40.00	11.76	14.63	49.87
MIST [28]	90.20	82.50	80.30	98.60	48.50	87.40	98.30	66.50	14.60	35.80	70.30
PCIR [30]	100.00	93.06	64.10	99.32	64.79	79.25	89.69	62.96	13.26	52.22	71.87
MIG [27]	97.79	90.26	87.18	98.65	54.93	64.15	100.00	74.07	12.96	21.57	70.16
TCA [29]	96.91	91.78	95.13	88.65	66.90	62.83	95.98	54.18	19.63	55.50	72.76
SAE [24]	97.06	91.67	87.81	98.65	40.86	81.13	100.00	70.37	14.81	52.22	73.46
SPG [26]	98.06	92.67	70.08	99.65	51.86	80.12	96.20	72.44	12.99	60.02	73.41
PISLM [18]	94.40	86.60	68.50	97.80	69.80	87.50	100.00	68.60	16.00	56.60	74.30
SGPLM [20]	98.20	93.80	89.30	99.10	50.20	88.90	100.00	71.00	12.30	51.20	75.40
AE-IS [25]	98.30	94.20	72.40	100.00	56.80	83.60	98.40	76.80	18.20	62.40	76.10
CICR(ours)	93.34	86.27	80.92	92.58	76.78	91.68	100.00	84.43	17.85	75.95	79.98

DIOR dataset. Note that each ICR branch is accompanied by a BBR branch; therefore, K also denotes the number of BBR branches.

As shown in Fig. 3, our method obtains the highest mAP when $K = 3$, and the mAP is slightly decreased when $K = 4$. Consequently, K is set to 3 in this article.

E. Ablation Study

1) Ablation Study of Feature Similarity: The ablation study of FS is conducted on the DIOR dataset to evaluate its effectiveness. As shown in Table I, the mAP and CorLoc of CPSL can be improved by using FS.

2) Ablation Study of CPSL and CMIL: To evaluate the effectiveness of CPSL and IMIL, the ablation studies are conducted on the more challenging DIOR [47] dataset. Similar to many WSOD methods [20], [21], [22], [28], the combination of OICR and BBR branches is adopted as the baseline of our CICR model to enhance the capability of object localization.

As shown in Table II, compared with the baseline model with one-hot label, the mAP (CorLoc) of baseline + CPSL (the traditional one-hot labels are replaced by the proposed CPSL) is improved by 4.49% (5.44%), the mAP (CorLoc) of baseline + IMIL is improved by 7.01% (9.12%), and the mAP (CorLoc) of baseline + CPSL + IMIL is improved by 10.96% (14.96%), which demonstrates the effectiveness of CPSL, IMIL, and their combination.

3) Ablation Study of IDS: The ablation study of IDS is conducted on the DIOR dataset to evaluate its effectiveness. As shown in Table III, the mAP and CorLoc of our model can be improved by using IDS.

F. Quantitative Comparison With Popular Methods

As shown in Tables IV–VII, to validate the overall capability of our CICR model, it is quantitatively compared with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.



Fig. 4. Qualitative comparisons with advanced methods on two RSI datasets. (b)–(d) Selected from the NWPU VHR-10.v2 dataset. (a), (e), and (f) Selected from the DIOR dataset.

WSDDN [16], OICR [17], PCL [32], the min-entropy latent model (MELM) [51], MIST [28], DCL [21], progressive contextual instance refinement (PCIR) [30], MIG [27], triple context-aware (TCA) [29], self-supervised adversarial and equivariant (SAE) [24], self-guided proposal generation (SPG) [26], PISLM [18], SGPLM [20], RINet [23], and attention erasing and instance sampling (AE-IS) [25] on two public RSI datasets, respectively.

As shown in Table IV (Table V), compared with WSDDN, OICR, PCL, MELM, MIST, DCL, PCIR, MIG, TCA, SAE, SPG, PISLM, SGPLM, RINet, and AE-IS, the mAP (CorLoc) of our CICR model is improved by 38.68% (44.74%), 39.28% (39.97%), 34.39% (34.92%), 31.51% (30.11%), 22.28% (9.68%), 21.69%, 18.83% (8.11%), 17.85% (9.82%), 14.98% (7.22%), 13.08% (6.52%), 11.00% (6.57%), 10.00% (5.68%), 8.60% (4.58%), 3.40%, and 1.80% (3.88%), respectively,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISONS WITH POPULAR WORKS ON THE DIOR DATASET IN TERMS OF MAP (%)

Method	Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf field
Fast R-CNN [3]	44.17	66.79	66.96	60.49	15.56	72.28	51.95	65.87	44.76	72.11
Faster R-CNN [1]	50.28	62.60	66.04	80.88	28.80	68.17	47.26	58.51	48.06	60.44
WSDDN [16]	9.06	39.68	37.81	20.16	0.25	12.28	0.57	0.65	11.88	4.90
OICR [17]	8.70	28.26	44.05	18.22	1.30	20.15	0.09	0.65	29.89	13.80
PCL [32]	21.52	35.19	59.80	23.49	2.95	43.71	0.12	0.90	1.49	2.88
MELM [51]	28.14	3.23	62.51	28.72	0.06	62.51	0.21	13.09	28.39	15.15
MIST [28]	32.01	39.87	62.71	28.97	7.46	12.87	0.31	5.14	17.38	51.02
DCL [21]	20.89	22.70	54.21	11.50	6.03	61.01	0.09	1.07	31.01	30.87
PCIR [30]	30.37	36.06	54.22	26.60	9.09	58.59	0.22	9.65	36.18	32.59
MIG [27]	22.20	52.57	62.76	25.78	8.47	67.42	0.66	8.85	28.71	57.28
TCA [29]	25.13	30.84	62.92	40.00	4.13	67.78	8.07	23.80	29.89	22.34
SAE [24]	20.57	62.41	62.65	23.54	7.59	64.62	0.22	34.52	30.62	55.38
SPG [26]	31.32	36.6	62.79	29.10	6.08	62.66	0.31	15.00	30.10	35.00
PISLM [18]	29.10	49.80	70.90	41.40	7.20	45.50	0.20	35.40	36.80	60.80
SGPLM [20]	39.10	64.60	64.40	26.90	6.30	62.30	0.90	12.20	26.30	55.30
RINet [23]	26.20	57.40	62.70	25.10	9.90	69.20	1.40	13.30	36.20	51.40
AE-IS [25]	31.80	50.90	63.20	29.40	8.90	68.70	1.30	15.10	35.50	51.60
CICR(ours)	36.42	63.70	60.43	32.29	9.19	70.29	1.89	14.96	36.78	53.16

Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill	mAP
Fast R-CNN [3]	62.93	46.18	38.03	32.13	70.98	35.04	58.27	37.91	19.20	38.10	49.98
Faster R-CNN [1]	67.00	43.86	46.87	58.48	52.37	42.35	79.52	48.02	34.77	65.44	55.48
WSDDN [16]	42.53	4.66	1.06	0.70	63.03	3.95	6.06	0.51	4.55	1.14	13.26
OICR [17]	57.39	10.66	11.06	9.09	59.29	7.10	0.68	0.14	9.09	0.41	16.50
PCL [32]	56.36	16.76	11.05	9.09	57.62	9.09	2.47	0.12	4.55	4.5	18.19
MELM [51]	41.05	26.12	0.43	9.09	8.58	15.02	20.57	9.81	0.04	0.53	18.66
MIST [28]	49.48	5.36	12.24	29.43	35.53	25.36	0.81	4.59	22.22	0.80	22.18
DCL [21]	56.45	5.05	2.65	9.09	63.65	9.09	10.36	0.02	7.27	0.79	20.19
PCIR [30]	58.51	8.60	21.63	12.09	64.28	9.09	13.62	0.30	9.09	7.52	24.92
MIG [27]	47.73	23.77	0.77	6.42	54.13	13.15	4.12	14.76	0.23	2.43	25.11
TCA [29]	53.85	24.84	11.06	9.09	46.40	13.74	30.98	1.47	9.09	1.00	25.82
SAE [24]	52.70	17.57	6.85	9.09	51.59	15.43	1.69	14.44	1.41	9.16	27.10
SPG [26]	48.02	27.11	12.00	10.02	60.04	15.10	21.00	9.92	3.15	0.06	25.77
PISLM [18]	48.50	14.00	25.10	18.50	48.90	11.70	11.90	3.50	11.30	1.70	28.60
SGPLM [20]	60.60	9.40	23.10	13.40	57.40	17.70	1.50	14.00	11.50	3.50	28.50
RINet [23]	53.90	28.60	4.80	9.10	52.70	15.80	20.60	12.90	9.10	4.70	28.30
AE-IS [25]	52.30	28.80	13.30	11.20	56.90	16.30	22.40	14.00	8.00	2.60	29.10
CICR(ours)	55.87	26.86	24.49	17.88	55.19	10.41	22.85	15.79	9.98	3.53	31.10

which shows the best performance of our model on the NWPU VHR-10.v2 dataset.

As shown in Table VI (Table VII), compared with WSDDN, OICR, PCL, MELM, MIST, DCL, PCIR, MIG, TCA, SAE, SPG, PISLM, SGPLM, RINet, and AE-IS, the mAP (CorLoc) of our CICR model is improved by 17.84% (25.41%), 14.60% (23.08%), 12.91% (16.33%), 12.44% (14.51%), 8.92% (14.25%), 10.91%, 6.18% (11.75%), 5.99% (11.05%), 5.28% (9.44%), 4.00% (8.43%), 5.33% (9.55%), 2.50% (4.65%), 2.60% (4.65%), 2.80% (5.05%), and 2.00% (1.95%), respectively, which shows the best performance of our model on the DIOR dataset.

Furthermore, as shown in Tables IV and VI, the overall capability of our CICR model is nearly comparable with two FSOD methods on the NWPU VHR-10.v2 dataset, and it further narrows the gap with FSOD methods on the DIOR dataset.

The reasons for our excellent performance include two aspects. On the one hand, the proposed CPSL can alleviate the misclassification problem attributed to that the neighboring instances are assigned the same label as the seed instance. As shown in Fig. 4(b), the SGPLM assigns a label of the tennis court to a basketball court; it is obvious that the wrong label is coming from the neighboring tennis court. However, our method can avoid the mistake because of the CPSL. On the other hand, the proposed IMIL scheme can handle the misclassification problem attributed to that the prediction errors of each ICR branch are accumulated with the iterative refinement of ICR branches. As shown in Fig. 4(d), some complex background regions are mistakenly recognized as ships or bridges, which have not existed in the RSI; meanwhile, our method can avoid the mistake because the proposed IMIL scheme uses the image-level labels as the second-level supervision of each ICR branch to suppress the misclassification arising from the object categories not existed in the RSI.

G. Qualitative Comparison With Advanced Methods

To further justify the superior performance of our CICR model, it is subjectively compared with two classical WSOD methods, i.e., OICR and MIST, and three competitive WSOD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
COMPARISONS WITH POPULAR WORKS ON THE DIOR DATASET IN TERMS OF CORLOC (%)

Method	Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf field
WSDDN [16]	5.72	59.88	94.24	55.94	4.92	23.40	1.03	6.79	44.52	12.75
OICR [17]	15.98	51.45	94.77	55.79	3.55	23.89	0.00	4.82	56.68	22.42
PCL [32]	61.14	46.86	95.39	63.61	7.32	95.07	0.21	5.71	5.14	50.77
MELM [51]	76.98	28.94	92.66	63.01	13.00	90.09	0.21	16.96	37.88	44.62
MIST [28]	91.60	53.20	93.50	66.30	10.80	30.70	1.50	14.03	35.20	47.50
PCIR [30]	93.10	45.60	95.50	68.30	3.60	92.10	0.20	5.40	58.40	47.50
MIG [27]	76.98	46.86	95.39	63.61	23.00	95.07	0.21	16.96	57.88	50.77
TCA [29]	81.58	51.33	96.17	73.45	5.03	94.69	15.89	32.79	45.95	48.56
SAE [24]	91.20	69.37	95.48	67.52	18.88	97.78	0.21	70.54	54.32	51.43
SPG [26]	80.48	32.04	98.68	65.00	15.20	96.08	22.52	16.99	46.08	50.96
PISLM [18]	85.50	68.90	96.80	75.80	11.60	94.70	0.80	67.50	60.50	46.50
SGPLM [20]	92.20	58.30	97.80	74.20	16.20	95.20	0.30	51.30	56.20	52.30
RINet [23]	92.70	80.90	92.70	69.50	8.60	90.10	0.20	71.30	62.00	65.50
AE-IS [25]	91.40	78.60	96.10	68.80	16.00	92.30	22.80	68.90	60.60	62.70
CICR(ours)	92.89	81.68	98.39	67.78	17.47	93.84	1.98	72.64	62.82	68.46

Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill	CorLoc
WSDDN [16]	89.90	5.45	10.00	22.96	98.54	79.61	15.06	3.45	11.56	3.22	32.44
OICR [17]	91.41	18.18	18.70	31.80	98.28	81.29	7.45	1.22	15.83	1.98	34.77
PCL [32]	89.39	42.12	19.78	37.94	97.93	80.65	13.77	0.20	10.50	6.94	41.52
MELM [51]	88.08	49.39	15.65	28.19	98.28	82.97	22.75	10.34	4.62	2.23	43.34
MIST [28]	87.10	38.60	23.40	50.70	80.50	89.20	22.40	11.50	22.20	2.40	43.60
PCIR [30]	88.60	15.80	5.20	39.50	98.10	85.60	13.40	56.50	9.70	0.60	46.10
MIG [27]	89.39	42.12	19.78	37.94	97.93	80.65	13.77	10.34	10.50	6.94	46.80
TCA [29]	85.26	38.91	20.17	30.63	84.59	91.46	56.28	3.79	10.45	1.25	48.41
SAE [24]	88.28	48.03	2.28	33.56	14.11	83.35	65.59	19.88	16.41	2.85	49.42
SPG [26]	89.18	49.45	22.00	35.16	98.61	90.04	32.56	12.73	9.98	2.34	48.30
PISLM [18]	75.20	50.50	28.30	39.70	92.60	77.00	55.10	10.10	20.90	5.60	53.20
SGPLM [20]	91.70	48.60	23.00	32.70	98.80	89.30	43.50	19.50	18.30	4.00	53.20
RINet [23]	85.10	51.40	15.70	44.60	98.60	80.30	14.80	22.70	6.90	2.60	52.80
AE-IS [25]	88.20	50.90	23.40	40.20	98.80	91.50	33.20	18.40	12.30	2.50	55.90
CICR(ours)	91.29	53.96	26.98	45.78	97.96	84.69	48.99	20.97	19.76	8.59	57.85

methods with available source codes, i.e., PISLM, SPGLM, and RINet.

As shown in Fig. 4(a), other WSOD methods assign a label of the tennis court to the basketball court; apparently, the mistake is attributed to the neighboring tennis court. Meanwhile, our CICR model can avoid this mistake by using the proposed CPSL. The similar case can be seen in Fig. 4(b). As shown in Fig. 4(c), other WSOD methods mistakenly identify a background region as a bridge that has not existed in the RSI; however, our model can handle this mistake by using the proposed IMIL scheme. The similar case can be seen in Fig. 4(d)–(f).

H. Analysis of Failure Detection Results

Although our CICR model has achieved better performance than popular WSOD models, it has some difficulties on some occasions. Some failure detection results are discussed as follows.

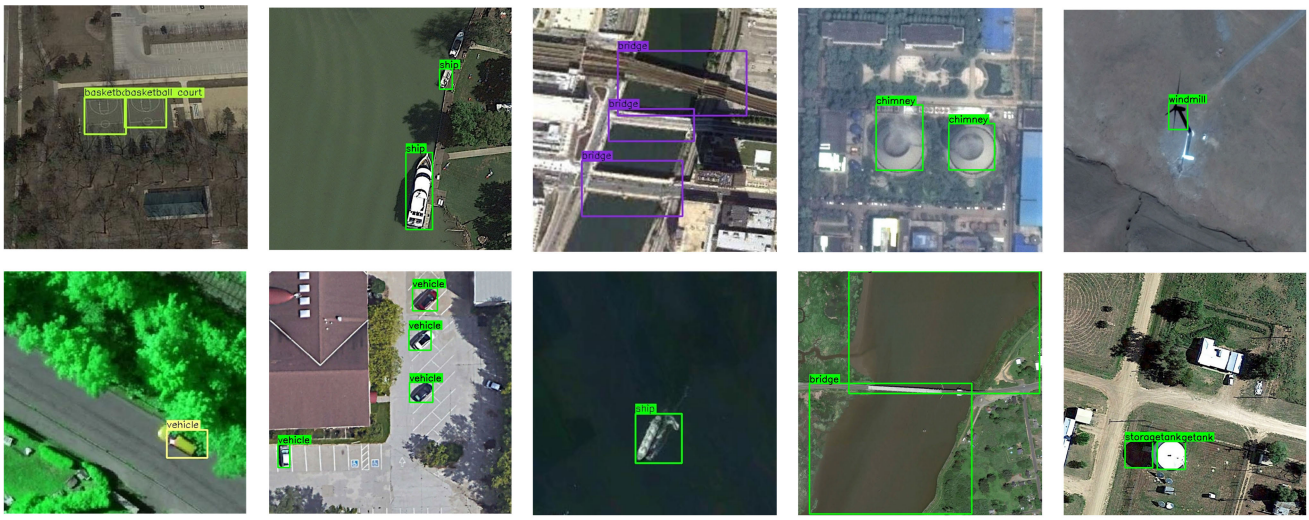
1) *Affected by Occlusion:* As shown in Fig. 5(a), the basketball courts and vehicle cannot be fully covered because of the occlusion. As shown in Fig. 5(b), a ship and a vehicle are missed for the same reason.
2) *Affected by Neighboring Background:* As shown in Fig. 5(c) and (d), some objects and their neighboring backgrounds are identified as the targets together because they have a strong contextual relationship with each other; in other words, the objects have coexisted with their neighboring backgrounds when only the image-level labels are provided.
3) *Affected by Shadow:* The shadows of some objects are very similar to the objects themselves under the condition of bird's-eye view imaging; therefore, as shown in Fig. 5(e), the shadows of windmill and storage tank are mistakenly detected as the targets.

I. Evaluation of Computational Cost

To evaluate the computational cost of our method, it is compared with two classical WSOD methods, i.e., OICR and MIST, and three advanced WSOD methods with available source codes, i.e., PISLM, SPGLM, and RINet.

As shown in Table VIII, the number of learnable parameters of six methods is similar, and the reasons for which are given as follows. First, the OICR model has been adopted as the basic framework of almost all WSOD methods since it was proposed in 2017; consequently, adding multiple ICR branches proposed by OICR has been used in popular methods. Second, adding multiple BBR branches is also adopted by popular WSOD methods because it can improve the performance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e)

Fig. 5. Illustration of some failure detection results on two RSI datasets. (a) and (c) Selected from the NWPU VHR-10.v2 dataset. (b), (d), and (e) Selected from the DIOR dataset.

TABLE VIII

COMPARISON WITH ADVANCED METHODS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF COMPUTATIONAL COST AND MAP

Method	Params (M)	Training time (H)	GFLOPS (G)	mAP (%)
OICR	134.60	3.25	286.23	34.52
MIST	135.02	6.50	290.43	51.52
PISLM	135.53	9.83	296.43	63.80
SGPLM	135.63	10.50	334.26	65.20
RINet	135.11	9.75	534.26	70.40
CICR(Ours)	135.08	8.51	294.37	73.10

localization. In summary, combining WSDDN with multiple ICR and BBR branches has been used as the baseline architecture by popular WOSD methods; therefore, the number of learnable parameters is similar.

As shown in Table VIII, the training time of our method is longer than the one of OICR and MIST, and the giga floating point operations (GFLOPs) of our method are slightly larger than the one of OICR and MIST; however, the mAP of our method is significantly larger than the one of OICR and MIST. On the whole, the comprehensive efficiency of our method is the best among the six methods.

V. CONCLUSION

A novel WSOD model named CICR is proposed in this article, and its main contributions include two aspects. On the one hand, a CPSL is proposed to handle the misclassification of neighboring instances caused by assigning a one-hot label to all of the seed instances and their neighboring instances. The CPSL of seed instances is defined according to the predicted class score of the upper ICR branch, and the CPSL of other instances is defined according to the spatial distance weighted FS between them and corresponding seed instances. The CPSL of all instances is used to directly train each ICR branch. On the other hand, the prediction mistake of each ICR branch is suppressed by the proposed IMIL scheme, which is accumulated with the propagation of supervisory information generated from the predicted class score of the upper ICR branch. The IMIL scheme uses the real image label to indirectly train each ICR branch. In addition, the affine transformations of the original RSI are incorporated into the above CPSL and IMIL to enhance the invariance of our CICR model. The ablation studies validate the effectiveness of CPSL, IMIL, and their combination. The quantitative and qualitative comparisons with popular works demonstrate the superiority of our CICR model on two RSI datasets.

The localization of our model relies on the proposals generated by the traditional SS algorithm of which the capability is limited; therefore, the future work of our method is to enhance the precision of localization by using advanced segmentation algorithms.

REFERENCES

[1] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.* vol. 39, no. 6, pp. 1137–1149, Jun. 2016.

[2] L. Li, X. Yao, X. Wang, D. Hong, G. Cheng, and J. Han, "Robust few-shot aerial image object detection via unbiased proposals filtration," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617011.

[3] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[4] G. Cheng, Q. Li, G. Wang, X. Xie, L. Min, and J. Han, "SFRNet: Fine-grained oriented object recognition via separate feature refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610510.

[5] X. Qian, B. Wu, G. Cheng, X. Yao, W. Wang, and J. Han, "Building a bridge of bounding box regression between oriented and horizontal object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605209.

[6] X. Xie, G. Cheng, Q. Li, S. Miao, K. Li, and J. Han, "Fewer is more: Efficient object detection in large aerial images," *Sci. China Inf. Sci.*, vol. 67, no. 1, 2024, Art. no. 112106.

[7] X. Xie, C. Lang, S. Miao, G. Cheng, K. Li, and J. Han, "Mutual-assistance learning for object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15171–15184, Dec. 2023.

[8] C. Liu et al., "High-spatial-resolution nighttime light dataset acquisition based on volunteered passenger aircraft remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 1001817.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.

[9] Y. Zhao, J. Qi, F. Korn, and X. Wang, "Scalable building height estimation from street scene images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3000718.

[10] Y. Yang, K. Yang, and Y. Chen, "Aerosol retrieval algorithm for Sentinel-2 images over complex urban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4107509.

[11] J. Chen, Y. Jiang, L. Luo, and W. Gong, "ASF-net: Adaptive screening feature network for building footprint extraction from remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4706413.

[12] Q. An et al., "Range-max enhanced ultrawideband micro-Doppler signatures of behind-the-wall indoor human motions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5107219.

[13] B. G. Palm, F. M. Bayer, R. Machado, M. I. Pettersson, V. T. Vu, and R. J. Cintra, "Robust Rayleigh regression method for SAR image processing in presence of outliers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5213412.

[14] W. Zhao et al., "Feature balance for fine-grained object classification in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620413.

[15] X. Qian, Y. Zeng, W. Wang, and Q. Zhang, "Co-saliency detection guided by group weakly supervised learning," *IEEE Trans. Multimedia*, vol. 25, pp. 1810–1818, 2023.

[16] H. Bilen and A. Vedaldi, "Weakly supervised deep detection networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2846–2854.

[17] P. Tang, X. Wang, X. Bai, and W. Liu, "Multiple instance detection network with online instance classifier refinement," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2843–2851.

[18] X. Qian, Y. Huo, G. Cheng, C. Gao, X. Yao, and W. Wang, "Mining high-quality pseudoinstance soft labels for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607615.

[19] X. Qian et al., "Multiscale image splitting based feature enhancement and instance difficulty aware training for weakly supervised object detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 7497–7506, 2023.

[20] X. Qian, C. Li, W. Wang, X. Yao, and G. Cheng, "Semantic segmentation guided pseudo label mining and instance re-detection for weakly supervised object detection in remote sensing images," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 119, May 2023, Art. no. 103301.

[21] X. Yao, X. Feng, J. Han, G. Cheng, and L. Guo, "Automatic weakly supervised object detection from high spatial resolution remote sensing images via dynamic curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 675–685, Jan. 2021.

[22] J. Seo, W. Bae, D. J. Sutherland, J. Noh, and D. Kim, "Object discovery via contrastive learning for weakly supervised object detection," in *Computer Vision—ECCV 2022*. Cham, Switzerland: Springer Nature, 2022, pp. 312–329.

[23] X. Feng, X. Yao, G. Cheng, and J. Han, "Weakly supervised rotation-invariant aerial object detection network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14126–14135.

[24] X. Feng, X. Yao, G. Cheng, J. Han, and J. Han, "SAENet: Self-supervised adversarial and equivariant network for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5610411.

[25] X. Xie, G. Cheng, X. Feng, X. Yao, X. Qian, and J. Han, "Attention erasing and instance sampling for weakly supervised object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600910.

[26] G. Cheng, X. Xie, W. Chen, X. Feng, X. Yao, and J. Han, "Self-guided proposal generation for weakly supervised object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625311.

[27] B. Wang, Y. Zhao, and X. Li, "Multiple instance graph learning for weakly supervised remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613112.

[28] Z. Ren et al., "Instance-aware, context-focused, and memory-efficient weakly supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 10598–10607.

[29] X. Feng, J. Han, X. Yao, and G. Cheng, "TCANet: Triple context-aware network for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6946–6955, Aug. 2021.

[30] X. Feng, J. Han, X. Yao, and G. Cheng, "Progressive contextual instance refinement for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8002–8012, Nov. 2020.

[31] Z. Huang, Y. Zou, B. V. K. V. Kumar, and D. Huang, "Comprehensive attention self-distillation for weakly-supervised object detection," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Red Hook, NY, USA: Curran Associates, 2020, pp. 16797–16807. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/file/c3535febaff29fcb7c0d20cbe94391c7-Paper.pdf

[32] P. Tang et al., "PCL: Proposal cluster learning for weakly supervised object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 1, pp. 176–191, Jan. 2020.

[33] J. Hosang, R. Benenson, and B. Schiele, "Learning non-maximum suppression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4507–4515.

[34] J. R. R. Uijlings, K. E. A. van de Sande, T. Gevers, and A. W. M. Smeulders, "Selective search for object recognition," *Int. J. Comput. Vis.*, vol. 104, no. 2, pp. 154–171, Sep. 2013.

[35] Z. Chen, Z. Fu, R. Jiang, Y. Chen, and X.-S. Hua, "SLV: Spatial likelihood voting for weakly supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12992–13001.

[36] Z. Zeng, B. Liu, J. Fu, H. Chao, and L. Zhang, "WSOD2: Learning bottom-up and top-down objectness distillation for weakly-supervised object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8291–8299.

[37] Z. Wu, J. Wen, Y. Xu, J. Yang, and D. Zhang, "Multiple instance detection networks with adaptive instance refinement," *IEEE Trans. Multimedia*, vol. 25, pp. 267–279, 2023.

[38] B. Liu et al., "Utilizing the instability in weakly supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jun. 2019, pp. 1–10.

[39] Y. Xu, C. Zhou, X. Yu, B. Xiao, and Y. Yang, "Pyramidal multiple instance detection network with mask guided self-correction for weakly supervised object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 3029–3040, 2021.

[40] X. Qian et al., "Incorporating the completeness and difficulty of proposals into weakly supervised object detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1902–1911, 2022.

[41] C. Lin, S. Wang, D. Xu, Y. Lu, and W. Zhang, "Object instance mining for weakly supervised object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 11482–11489.

[42] G. Yan et al., "C-MIDN: Coupled multiple instance detection network with segmentation guidance for weakly supervised object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9833–9842.

[43] D. Biswas and J. Tešić, "Unsupervised domain adaptation with debiased contrastive learning and support-set guided pseudolabeling for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3197–3210, 2024.

[44] D. Biswas and J. Tešic, "Small object difficulty (SOD) modeling for objects detection in satellite images," in *Proc. 14th Int. Conf. Comput. Intell. Commun. Netw. (CICN)*, Dec. 2022, pp. 125–130.

[45] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[46] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[47] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

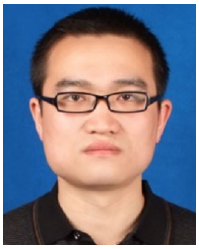
[48] T. Deselaers, B. Alexe, and V. Ferrari, "Weakly supervised localization and learning with generic knowledge," *Int. J. Comput. Vis.*, vol. 100, no. 3, pp. 275–293, Dec. 2012.

[49] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[50] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, pp. 1097–1105.

[51] F. Wan, P. Wei, J. Jiao, Z. Han, and Q. Ye, "Min-entropy latent model for weakly supervised object detection," in *Proc. IEEE CVPR*, Jun. 2018, pp. 1297–1306.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.



Xiaoliang Qian (Member, IEEE) received the Ph.D. degree from the School of Automation, Northwestern Polytechnical University, Xi'an, China, in 2013.

He is currently an Associate Professor with the College of Electrical and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou, China. He has authored over 60 papers in international journals and conferences. His research interests include computer vision, deep learning, and remote sensing image understanding.

Dr. Qian serves as a peer reviewer for IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), and IEEE Transactions on Systems Man, Cybernetics: Systems (TSMCA).



Chenhao Wang received the B.S. degree from Huanghe Jiaotong University, Jiaozuo, China, in 2021. He is currently pursuing the M.S. degree with Zhengzhou University of Light Industry, Zhengzhou, China.

His research interests include object detection in remote sensing images and deep learning.



Wei Wang received the Ph.D. degree from Concordia University, Montreal, QC, Canada, in 2002.

He is currently a Professor with the School of Electrical and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou, China. His research interests include artificial intelligence, remote sensing image understanding, computer vision, and tactile sensing.



Xiwen Yao (Member, IEEE) received the B.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 2010 and 2016, respectively.

He is currently an Associate Professor with Northwestern Polytechnical University. His research interests include computer vision and remote sensing image processing, with an emphasis on fine-grained image classification and object detection.



Gong Cheng (Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2007, and the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, in 2010 and 2013, respectively.

He is currently a Professor with Northwestern Polytechnical University. His research interests include computer vision, pattern recognition, and remote sensing image understanding.

Dr. Cheng is also an Associate Editor of the *IEEE Geoscience and Remote Sensing Magazine* and a Guest Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:43:20 UTC from IEEE Xplore. Restrictions apply.